

TABLE II
COMPARISON BETWEEN DIFFERENT SINGLE-BOARD MICROCOMPUTERS

Name	SoC	Architecture	Supported OS	Onboard Storage (MB)	External Storage	HDMI port	Other video Port	Ethernet Port	Wi-Fi	Standard USB (2.0/3.0) ports	Figure
Humming-Board-i2eX	Freescale i.MX6 Dual	ARM Cortex-A9	Linux, Android	—	microSD mSATA	Yes	—	GbE	—	USB 2.0	
Raspberry Pi's B+	Broadcom BCM2835	ARM11	Linux (Raspbian), Android	—	microSD	Yes	Composite	10/100	—	USB 2.0	
Banana Pi M3	Allwinner A83T	ARM Cortex-A7	Linux, Android	—	microSD SATA 2.0	Yes	Composite	GbE	a/b/g/n	USB 2.0	
Intel's Galileo Gen2 development board	Intel Quark SoC X1000	x86 Quark	Linux, Windows	8MB flash + 8kB EEPROM	SD	—	—	10/100	—	USB 2.0	
NanoPC-T1	Samsung Exynos 4 (4412)	ARM Cortex-A9	Linux	8GB flash	SD	Yes	—	10/100	—	USB 2.0	
ODROID-C1	Amlogic S805	ARM Cortex-A5	Linux, Android	eMMC module optional	microSD	Yes	—	10/100/1000	—	USB 2.0	
pcDuino-3Nano	Allwinner A20	ARM Cortex-A7	Linux, Android	4GB flash	microSD	Yes	—	GbE	—	USB 2.0	
Cubieboard 4	Allwinner A80	ARM Cortex-A15x4	Linux, Android	8GB eMMC	SD	Yes	VGA	GbE	b/g/n	USB 2.0 USB 3.0	
BeagleBone Black	Ti Sitara AM335x	ARM Cortex-A8	Linux, Android, Windows	4GB eMMC	microSD	Yes	—	10/100	—	USB 2.0	
OLinuXino A20 LIME2	Allwinner A20	ARM Cortex-A7	Linux, Android	4GB flash optional	microSD	Yes	—	1000	—	USB 2.0	
Radxa Rock	Rockchip RK3188	ARM Cortex-A9	Linux, Android	8GB flash	microSD	Yes	AV-output	10/100	b/g/n	USB 2.0	

Windows CE.

Programming/debugging tools are also often free and open source, such as those based on Eclipse IDE. Other tools that are tailored to a specific processor often used by professionals include ARM's DS-5 or vendor-specific tools such as Freescale's CodeWarrior or Texas Instruments' Code Composer Studio.

While SBCs can be used for most

purposes, many have originally been designed for a specific purpose or application. A perfect example of this is Raspberry Pi, which was developed as an educational tool to help encourage and strengthen students' programming skills.

BeagleBoard and BeagleBone were also developed to help educate and promote the benefits and usage of open source hardware and soft-

ware in embedded computing.

Atmel's SAM5D3 Xplained has been designed for rapid prototyping development, and RIoTboard focuses on Android development to enable the development of the Internet of Things (IoT).

Other well-known boards include PandaBoard, OlinuXino as well as a whole host of Allwinner ARM system on chip (SoC) based SBCs.